I realised that my daughter was strong enough to console me, so she should be able to keep my wife and son strong too.

The car appeared to have slowed down. And I could not wait to see my daughter.

PSLV TAKES SHAPE

'I will cancel the order.'

Even as we kept sourcing materials from abroad to put together the Vikas engine for the first PSLV flight, indigenisation efforts were on for further flights. We appeared to be well on our way to self-sufficiency when it came to UDMH fuel that our team was now producing at the Andhra Sugar facility. But development of some materials, especially alloys like HS-25 used for making the bell-shaped nozzle of Vikas, was proving tricky.

Getting MIDHANI on board helped us rope in other big public sector units collaborate with ISRO for the Vikas project. Hindustan Aeronautics Limited (HAL), Bangalore took up the job of fabricating the thrust frame, tankages and inter-stage structures. We still had a lot of components of the engine to be made.

One of the early challenges of rocketry the world over faced was the rocket's tendency to spin on its axis. This was kept under check through a roll control system. The changes in direction were to be effected by a process called gimballing, wherein the nozzle moves to change the direction of the thrust, and thereby that of the rocket.

A team led by ISRO's components group head Sreedharan Das along with ISRO engineers E S Joseph, Radhakrishnan, Baby T Mathew, Ram Jogi and Thomas Jayakumar laboured to perfect the flow components of the control system. This included the Group Organ de Commande (GOC), essentially a plate that holds together several valves and regulators, which